

Contact

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www.linkedin.com/in/james-wanjiku
(LinkedIn)
garnet-owl.github.io/ (Portfolio)

Top Skills

Python (Programming Language)
Natural Language Processing (NLP)
Computer Vision

Certifications

Natural Language Processing
Specialization
Machine Learning Specialization
Introduction to Large Language
Models
Mathematics for Machine Learning

Publications

Towards Reliable Instruction-
Following in LLM Multi-Turn
Workflows: An Empirical Comparison
of Prompting and Control
Architectures

James Wanjiku

AI/ML Engineer, Team Lead at AAI Labs | Voice AI, LLM systems,
computer vision | Python, GCP
Nairobi County, Kenya

Summary

I build and ship AI systems that run in production, mostly voice agents and LLMpipelines. Right now I lead the engineering team on an autonomous B2B voice calling platform that handles thousands of concurrent calls across multiple languages, with an assisted mode where a human operator can step into a live call. I run delivery across both the voice/AI services and the backend and web app, and I still write and deploy code. Before that I built the georeferencing and object-tracking pipeline for a 5G-connected drone system: live video and telemetry in, detection and tracking, then every detection resolved to real GPS coordinates from the drone's position and camera orientation. The work was presented to and accepted by an EUcommittee. I also lead technical concept and writing on AAI Labs' EU research grant proposals, and coordinate consortium partners across Europe. On the research side I led a three-person team on a study of how reliably LLMs follow instructions across multi-turn workflows. The paper was accepted at ICIST 2026 in Kaunas. Outside work I maintain Taura, an open-source Kikuyu-English translation engine. Kikuyu has nine to ten million speakers and almost no machine translation support, which seemed worth fixing. Nairobi based, working remotely with a Lithuanian team.

Experience

AAI Labs

2 years 2 months

AI/ML Engineer, Team Lead

July 2026 - Present (2 months)

Nairobi County, Kenya

Lead engineering on an autonomous B2B voice calling platform for a European

client. The system places outbound calls, holds the conversation itself using

speech recognition, text to speech, and an LLM, and brings in a human operator

when a call needs one.

- Delivered parallel calling, taking the platform from calls placed one at a time to thousands running concurrently across multiple languages.
- Validated concurrency end to end against the telephony provider to establish how many simultaneous live calls the platform sustains before audio quality degrades.
- Shipped assisted calling, where a human operator can join or take over a call while it is still in progress.
- Built automatic language detection into the call pipeline, so the agent switches language to match the person who answered.
- Work across the real-time voice services and the web application the client's sales team uses, and run the AI services in production on Google Cloud.

AI/ML Engineer

July 2024 - Present (2 years 2 months)

Nairobi County, Kenya

Delivery and research across voice AI, computer vision, EU research proposals, and LLM evaluation.

- Built the georeferencing and object-tracking pipeline for a 5G-connected drone system. Live video and telemetry in, YOLO detection and multi-object tracking, then every detection ray-cast to GPS coordinates using the drone's position and gimbal orientation. Presented to and accepted by an EU committee.
- Lead the technical concept and proposal writing for AAI Labs' EU research grant applications, and coordinate consortium partners across Europe.
- Led a three-person team on an empirical study of instruction-following in multi-turn LLM workflows, comparing prompting against explicit control architectures. Accepted at ICIST 2026, Kaunas.
- Built an NLP system for a public-sector client that compares large volumes of funding applications and surfaces likely duplicates for human review.
- Built a recommendation service pairing activities to supervisor and child pairs from historical records, and backend work on a public-sector prototype for detecting duplicated funding requests.

JayBee Internet

Network Administrator

August 2023 - May 2024 (10 months)

Kenya

Maintained network infrastructure and user support before moving into AI engineering.

Education

University of Embu

Bachelor of Science - BS, Computer Science · (August 2018 - April 2023)